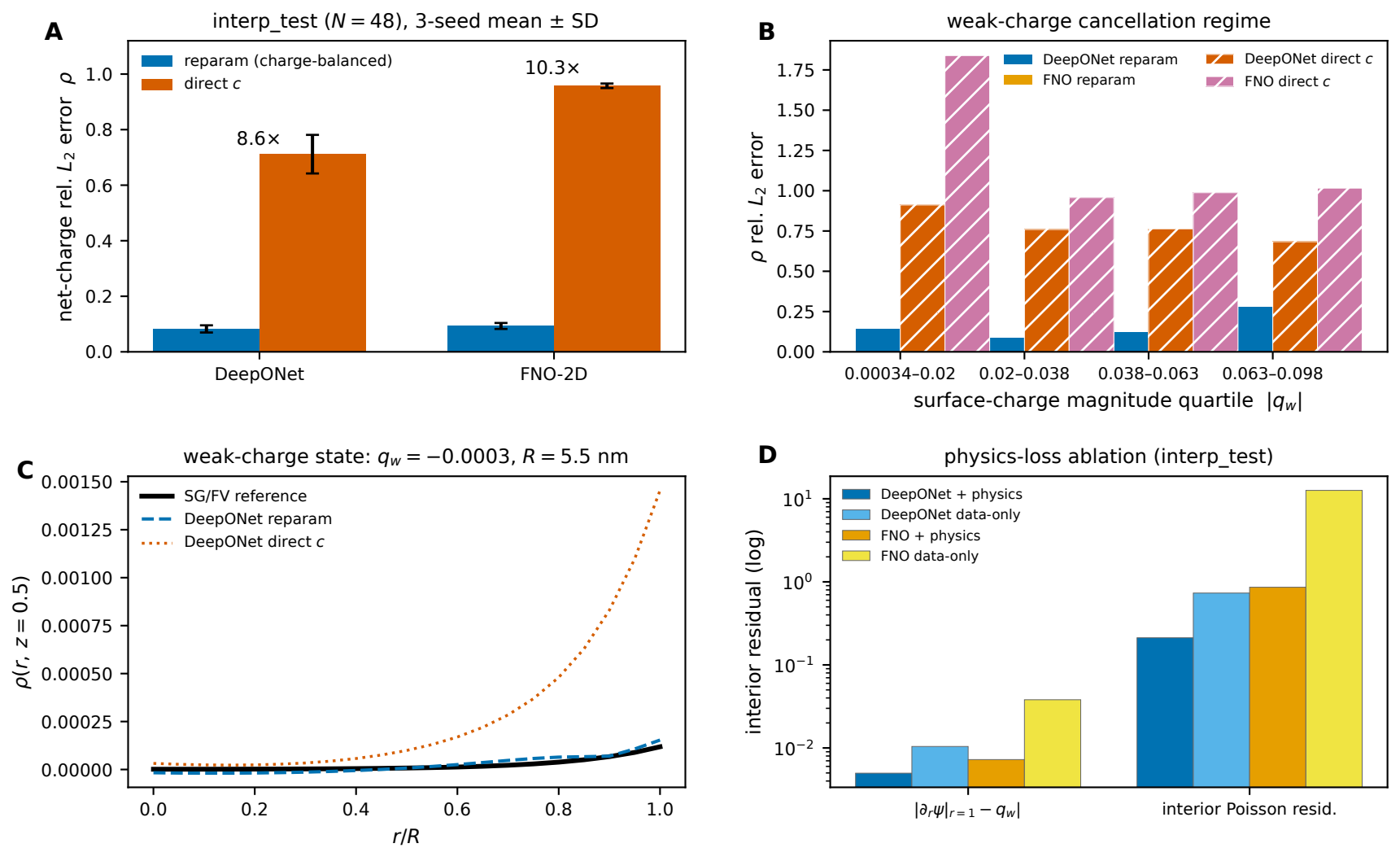
A
interp_test ($N = 48$), 3-seed mean ± SD
reparam (charge-balanced)
direct c
8.6×
10.3×
net-charge rel. L_2 error ρ
DeepONet
FNO-2D
B
weak-charge cancellation regime
DeepONet reparam
FNO reparam
DeepONet direct c
FNO direct c
ρ rel. L_2 error
0.00034–0.02
0.02–0.038
0.038–0.063
0.063–0.098
surface-charge magnitude quartile $|q_w|$
C
weak-charge state: $q_w = -0.0003$, $R = 5.5$ nm
SG/FV reference
DeepONet reparam
DeepONet direct c
$\rho(r,\ z = 0.5)$
r/R
D
physics-loss ablation (interp_test)
DeepONet + physics
DeepONet data-only
FNO + physics
FNO data-only
interior residual (log)
$|\partial_r \psi|_{r=1} - q_w|$
interior Poisson resid.